

\CurrentAudience: Class

PH24101 / Assignment - Module IV: Quantum Physics

(Dated: March 23, 2026)

We begin with clearly highlighting various (sub-)sections of the textbook which spans our syllabus for this module on physical optics.

Further, we present some of the *benchmark* question, mostly numerical, from Module IV to be discussed in the tutorials. *All the questions not marked by [!] are based directly on the classroom discussions.* It is expected that the students will solve these questions on their own, or in groups before the tutorial(s). At the same time, questions similar to the ones contained herein are likely to be asked in your exams. However, note that (i) same questions should NOT be expected in the exams, (ii) numerical questions beyond these could also be asked in the exams. In other words, this document is NOT a question bank of any kind. Therefore, one should aim to understand the underlying concepts well. **Solved examples from the textbook “Modern Physics” by Arthur Beiser are highly recommended.**

All corrections, comments, or suggestion should be emailed to
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Topics

TABLE I. Based on the book *Concepts of Modern Physics* by Arthur Beiser, McGraw Hill International Edition (Sixth Ed., ISBN: 0-07-244848-2, 0-07-115096-X).

Topic	Relevant (sub-)section from book
Planck's theory of black-body radiation,	Sec 2.2, 9.5, 9.6
Compton effect	Sec 2.7
Wave particle duality, De Broglie waves	Sec 3.1
Davisson and Germer's experiment	Sec 3.5
Uncertainty principle	Sec 3.7, 3.8, 3.9
Brief idea of Wave Packet, Wave Function and its physical interpretation	Sec 3.2, 3.4, 5.1, 5.2
Schrodinger equation in one-dimension, free particle, particle in an infinite square well	Sec 5.1 - 5.8

1. Calculate the de Broglie wavelength of (i) a 10 eV electron, and (ii) a 1.0MeV neutron. (Assume classical non-relativistic particles; Mass of neutron is 1.675×10^{-27} kg = 939.6 Mev/c² $\approx m_{\text{proton}}$.)

Comment: For relativistic particles, one can show that:

$$\lambda = \frac{hc}{\sqrt{E_K^2 + 2m_0E_Kc^2}}$$

where E_K is the kinetic energy.

2. An increase of 150 eV in the energy of an electron reduces its de Broglie wavelength to half of its earlier value. Calculate the initial de Broglie wavelength.
3. What is the potential through which a proton must be accelerated so that its de Broglie wavelength is 2.0 Å?
4. An electron has a de Broglie wavelength equal to that of a photon of red light having wavelength 620 nm. (i) Calculate the energy of the electron. (ii) What would its de Broglie wavelength be if the energy of the electron were equal to the energy of this photon?
5. **Compton effect.** X-rays of wavelength 10.0 pm are scattered through a target.
 - (a) Find the wavelength of the x-rays scattered through 45°.
 - (b) Find the maximum wavelength present in the scattered x-rays.
 - (c) Find the maximum kinetic energy of the recoil electrons.
6. What is the frequency of an x-ray photon whose momentum is 1.1×10^{-23} kg m/s²?
7. How much energy must a photon have if it is to have a momentum of 10 MeV proton?
8. An x-ray photon of initial frequency 3×10^{19} Hz collides with an electron and is scattered through 90°. Find its new frequency.
9. An x-ray photon with initial frequency of 1.5×10^{19} Hz emerges from a collision with an electron with a frequency of 1.2×10^{19} Hz. How much kinetic energy was imparted to the electron?

10. At what scattering angle will incident 100 keV x-rays leave a target with an energy of 90 keV?
11. A photon of frequency ν is scattered by an electron initially at rest. Verify that the maximum kinetic energy of the recoil electron is

$$K_{\max} = \frac{2h^2\nu^2}{m_0c^2} \frac{1}{1 + 2h\nu/m_0c^2}$$

12. **Davison-Germer expt.** What effect on the scattering angle in the Davisson-Germer experiment does increasing the electron energy have?
13. In the Davisson-Germer experiment, the energy of the electron entering the crystal increases, which reduces its de Broglie wavelength. Consider a beam of 54 eV electrons directed at a nickel target. The potential energy of an electron that enters the target changes by 26 eV. (a) Compare the electron speeds outside and inside the target. (b) Compare the respective de Broglie wavelengths.
14. Write down the (time-independent) Schrödinger equation for free particle and its solution.
15. **Particle in a box.** Obtain the energy values and wavefunctions of a quantum particle in a one-dimensional box. (Write down the Schrödinger equation for particle in a box and solve it). Plot the lowest two energy wavefunctions.
16. The lowest energy of a quantum particle trapped in a (unspecified) box is 1 eV. What are the next two higher energies the particle can have.
17. Find the energy levels of an electron in a box 0.1 nm wide.
18. Find the energy levels of a 10 g marble in a box 10 cm wide assuming the marble to be a quantum particle.

Comment: Using this example, convince yourself that quantum effects from a non-quantum object is actually negligible.

19. Obtain the expression for the energy levels (in MeV) of a neutron confined to a one-dimensional box 10^{-14} m wide. What is the neutron's minimum energy?

| *Comment: The diameter of an atomic nucleus is of this order of magnitude.*

20. **Uncertainty principle.** A measurement establishes the position of a proton with an accuracy of $\pm 10^{-11}$ m. Find the uncertainty in the proton's position 1 s later. Assume $v \ll c$.
21. A typical nucleus is about 5×10^{-15} m in radius. Use the uncertainty principle to place a lower bound on the energy an electron must have if it is to be part of a nucleus.
22. Same exercise as above (previous question) for a hydrogen atom, with $\Delta x = 5.3 \times 10^{-11}$ m, will lead to kinetic energy approximately equal to 3.4 eV.

| *Comment: Note: The kinetic energy of an electron in the lowest energy level of a hydrogen atom is actually 13.6 eV and can be obtained by solving the Schrödinger equation for the hydrogen atom.*

23. Discuss the Plank's theory of black-body radiation.

*Comment: **Concluding Remarks.** The students are **strongly recommended** to take a look at the relevant solved examples from the book *Concepts of Modern Physics* by Arthur Beiser, and also solve them explicitly.*

It is, in general, not possible to include all those examples here and discuss them explicitly. Based on the lectures, the students are expected to have sufficient skills to solve those problems. The relevant topics can be found from the course syllabus.